ICML 2022: Logistic Regression Is Only Half the Answer for Learning Halfspaces

A tight √OPT lower bound — and a two-step "logistic + perceptron" fix that restores near-optimal accuracy on any well-behaved distribution. From UIUC, MIT, Google Research, and CMU.

Logistic regression is probably the most widely deployed classifier in existence: one convex objective, one call, and it is nearly everyone's first move. Yet a basic question about it stayed unsettled for a long time — when the labels are corrupted by noise, how well does logistic regression actually learn?

This ICML 2022 paper trains that question on one of the oldest objects in learning theory: agnostically learning homogeneous halfspaces. A halfspace is just a linear classifier through the origin. "Agnostic" means we let an adversary flip an OPT fraction of the labels of an otherwise perfectly linearly separable dataset, so the best any linear classifier can hope for is a zero-one error of OPT — the error of that optimal halfspace itself. The goal is to get close to OPT on the corrupted data.

Before this work, the theory sat between two bounds with a wide gap in the middle. On one side, a lower bound says no convex surrogate loss can beat Ω(OPT). On the other, the best known upper bound only guarantees logistic regression reaches Õ(√OPT). Which bound reflects the truth for logistic regression? The paper closes the gap — and, along the way, hands over a surprisingly cheap fix.

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The setup: agnostically learning a halfspace

The problem is clean. We draw feature-label samples from an unknown distribution and want a linear classifier whose zero-one error is close to OPT, the smallest error achievable by any homogeneous halfspace. Equivalently, hand a perfectly separable dataset to an adversary who may flip an OPT fraction of its labels; we then try to recover the original decision boundary as closely as the corruption allows.

Logistic regression is the natural heuristic here — it minimizes a smooth convex loss, so it is easy to optimize and to analyze. The catch is its worst-case behavior, which is alarming: Ben-David et al. (2012) showed that on an adversarially constructed distribution, the logistic-risk minimizer can be wrong on nearly a 1−OPT fraction of examples. To rule out such pathologies, the field restricts attention to well-behaved distributions (isotropic log-concave ones, for instance), where much stronger guarantees hold. But even under that assumption, whether logistic regression achieves order OPT or order √OPT was open — and that gap is exactly what decides whether we can recommend it with confidence.

The bad news: logistic regression is stuck at √OPT — and it is tight

The paper's first result is a lower bound that settles which side of the gap is real: even on a well-behaved distribution, logistic regression cannot escape √OPT. The argument is not abstract — the authors build an explicit counterexample. The distribution Q below is two-dimensional, isotropic, and fully well-behaved, yet it reliably defeats logistic regression.

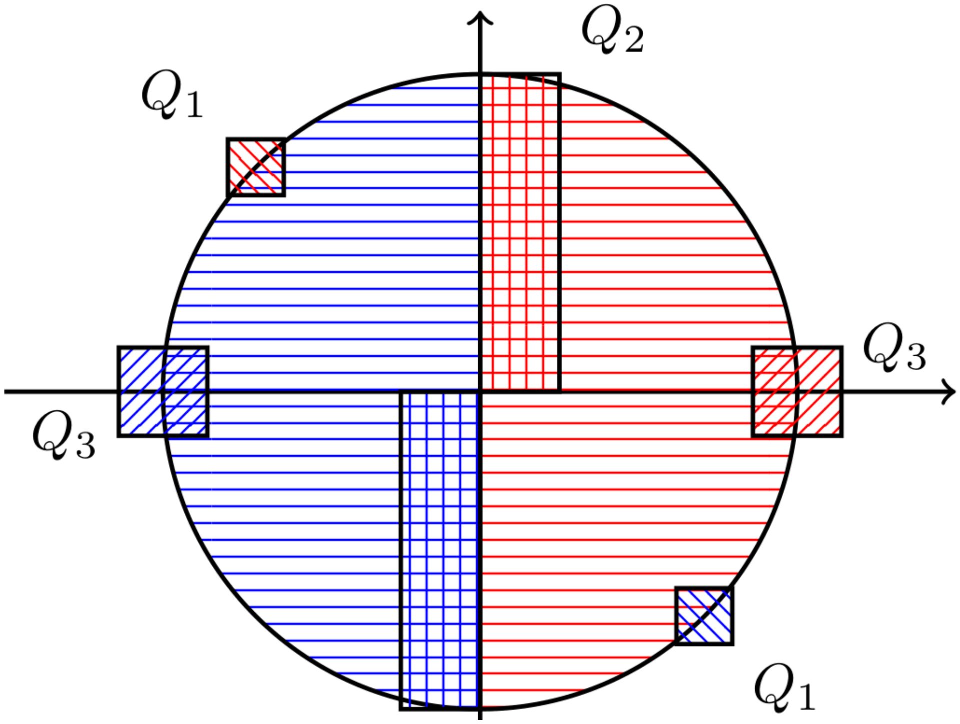

Figure 1. The counterexample distribution Q the paper constructs — two-dimensional, isotropic, and well-behaved. A background component Q_4 fills the unit disk, with labels set by the sign of x_1 (red = +1, blue = −1), so the truly optimal halfspace is the vertical axis. The small hatched squares Q_1, Q_2, Q_3 hold a carefully placed sliver of noisy examples: together they carry only an OPT fraction of the mass, yet they are enough to tilt the logistic-risk minimizer off the axis and force a zero-one error of order √OPT.

The whole construction is about leverage. The background Q_4 makes the distribution look tame, and the optimal boundary is plainly the vertical axis. But the little squares in the corners, though they carry almost no mass (only OPT of it), are positioned so precisely that the logistic loss over-weights them during optimization and settles on a direction pulled away from the axis. The authors prove that on this Q, the global minimizer w* of the logistic risk has zero-one error at least √OPT/(60π), for OPT small enough (the construction requires OPT ≤ 1/100). That √OPT exactly matches Frei et al.'s known Õ(√OPT) upper bound — so √OPT is the true rate for logistic regression on its own, and no analysis can do better. That is precisely what it means to close the gap.

When logistic regression already suffices: radial Lipschitzness

If the √OPT pathology comes from the density pulling unevenly at different radii, the natural question is what condition makes that pull vanish. The paper's answer is clean: radial Lipschitzness of the density. Intuitively, as long as the feature density changes gently along the radial direction, logistic regression is not dragged off course by a pocket of corner noise, and it reaches the near-optimal $\tilde{O}(\mathrm{OPT})$ rate on its own.

More precisely, the upper bound reads $O((1+C_\kappa)\cdot\mathrm{OPT})$, where the constant C_κ is an integral of the radial-Lipschitzness modulus: the smoother the density is along the radius, the smaller C_κ, and it vanishes entirely when the density is radially symmetric. The same upper-bound argument is also notably versatile — it recovers the earlier $\tilde{O}(\sqrt{\mathrm{OPT}})$ result as a special case, and it goes through for the hinge loss as well as the logistic loss. That last point is exactly what makes the two-phase algorithm below possible.

The cheap fix: logistic regression plus one perceptron step

Radial Lipschitzness is, after all, an extra assumption. The paper's most useful contribution is a two-phase algorithm that needs no such assumption and works for every well-behaved distribution — and it is barely more than "run logistic regression, then take one more step."

Phase one is logistic regression itself: projected gradient descent on the logistic risk inside a ball of radius $1/\sqrt{\varepsilon}$, returning a direction v. By the analysis above, v lands within angle $\tilde{O}(\sqrt{\mathrm{OPT}}+\varepsilon)$ of the true direction $\bar{u}$ — roughly right, but the zero-one error is still stuck at $\sqrt{\mathrm{OPT}}$. Phase two then runs projected stochastic gradient descent on the hinge loss, whose update rule is exactly the classical perceptron: starting from the warm start v and confined to a bounded domain, it keeps sharpening. This tightens the angle and drives the zero-one error down to order OPT.

The savings are the striking part. Prior algorithms that reached $O(\mathrm{OPT}+\varepsilon)$ had to solve $O(\log(1/\mathrm{OPT}))$ minimization problems while binary-searching for OPT; this method needs just two convex steps — logistic regression plus one perceptron phase. Its sample complexity also drops from the $\tilde{O}(d/\varepsilon^4)$ of earlier nonconvex approaches to $\tilde{O}(d/\varepsilon^2)$, a quadratic improvement.

Table 1. The two-phase algorithm versus prior nonconvex methods: the error rate matches, but the number of convex minimizations and the sample complexity both fall sharply.

Method	Zero-one error	Convex minimizations	Sample complexity
Prior nonconvex methods	$O(\mathrm{OPT} + \varepsilon)$	$O(\log(1/\mathrm{OPT}))$	$\tilde{O}(d/\varepsilon^4)$
This paper (two-phase)	$O(\mathrm{OPT} + \varepsilon)$	2	$\tilde{O}(d/\varepsilon^2)$

The rates at a glance

Laid side by side, the story of where logistic regression falls short and how to patch it becomes obvious: on its own it is stuck at $\sqrt{\mathrm{OPT}}$, but either adding radial Lipschitzness or appending one perceptron step brings it back to order OPT.

Table 2. Zero-one error rates across settings. OPT is the best error any linear classifier can achieve; on well-behaved distributions logistic regression only reaches $\sqrt{\mathrm{OPT}}$, a rate the paper's counterexample proves is tight.

Setting / method	Zero-one error rate
Best linear classifier (baseline)	OPT
Logistic regression, general well-behaved dist. (known upper bound)	$\tilde{O}(\sqrt{\mathrm{OPT}})$
Logistic regression on the counterexample Q (this	$\geq \sqrt{\mathrm{OPT}}/(60\pi)$

paper's lower bound)	
Logistic regression + radial Lipschitzness	Õ(OPT)
Two-phase algorithm, bounded distributions	O(OPT + ε)
Two-phase algorithm, sub-exponential distributions	O(OPT·ln(1/OPT) + ε)

Takeaways and boundaries

Two things are worth carrying away. First, for agnostically learning halfspaces, logistic regression on its own cannot beat √OPT, and that rate is tight — the paper corners it with an explicit well-behaved distribution. Second, the remedy is remarkably cheap: warm-start from the logistic solution, append one perceptron-style convex minimization, and the guarantee jumps to near-optimal Õ(OPT) on any well-behaved distribution, more simply and with better sample efficiency than any prior method. Along the way, the paper identifies radial Lipschitzness as the natural condition under which logistic regression already succeeds.

A note on scope: this is a theory paper. The results are theorems with explicit constants, not experimental curves, and every guarantee rests on the well-behaved family of distributions (isotropic, bounded or sub-exponential, satisfying soft-margin and regularity conditions). The distribution Q is a hand-crafted two-dimensional instance built to prove the lower bound, not real data. What the paper delivers is a sharp picture of how good logistic regression can be in principle, and the minimal step that closes the remaining gap — an intuition worth carrying back to practical noisy linear-classification tasks.